

Microlocal Morse Theory and Truncated Microsupport

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Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncated microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Truncated Microsupport

Definition 1.1. (i) Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if F satisfies the equivalent conditions in Prop.

(ii) aa

2 Results

Proposition 2.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. let $a < b$ be real numbers. We assume $d\psi(x) \notin SS_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}([-\infty, t])$. Then, restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j \leq k$.

Proof. First, we have

$$H^j(M_t; F) \cong H^j \text{R}\Gamma([-\infty, t[; \text{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin SS_k(F)$ for $a \leq \psi(x) < b$, and by the formula $SS_k(\text{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} SS_k(F)$, we have

$$(H^j \text{R}\Gamma_{[a, b[} \text{R}\psi_* F)_{\psi(x)} = 0$$

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for each $j \leq k$. Therefore, By the long exact sequence

$$\begin{aligned}
0 &\rightarrow H^0 \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \rightarrow H^0 \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \rightarrow H^0 \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\rightarrow H^1 \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \rightarrow H^1 \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \rightarrow H^1 \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\rightarrow \dots \\
&\rightarrow H^k \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\rightarrow H^{k+1} \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \rightarrow \dots,
\end{aligned}$$

we have isomorphisms

$$H^j \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \xrightarrow{\sim} H^j \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \quad (2.1)$$

for all $j \leq k$. Then we can conclude that

$$H^j(M_b ; F) \xrightarrow{\sim} H^j(M_a ; F)$$

by applying to the equality (2.1) the functor $\Gamma(]-\infty, b[; \cdot)$ and by calculate as follows:

$$\begin{aligned}
\Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty,b[}(\cdot)(]-\infty, b[; \mathrm{R}^j \psi_* F) \\
&\cong \Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F) \\
&\cong H^j(\psi^{-1}(]-\infty, b[) ; F) \\
&\cong H^j(M_b ; F)
\end{aligned}$$

and

$$\begin{aligned}
\Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty,a[}(\cdot)(]-\infty, b[; \mathrm{R}^j \psi_* F) \\
&\cong \Gamma(]-\infty, a[; \mathrm{R}^j \psi_* F) \\
&\cong H^j(\psi^{-1}(]-\infty, a[) ; F) \\
&\cong H^j(M_a ; F).
\end{aligned}$$

□

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